

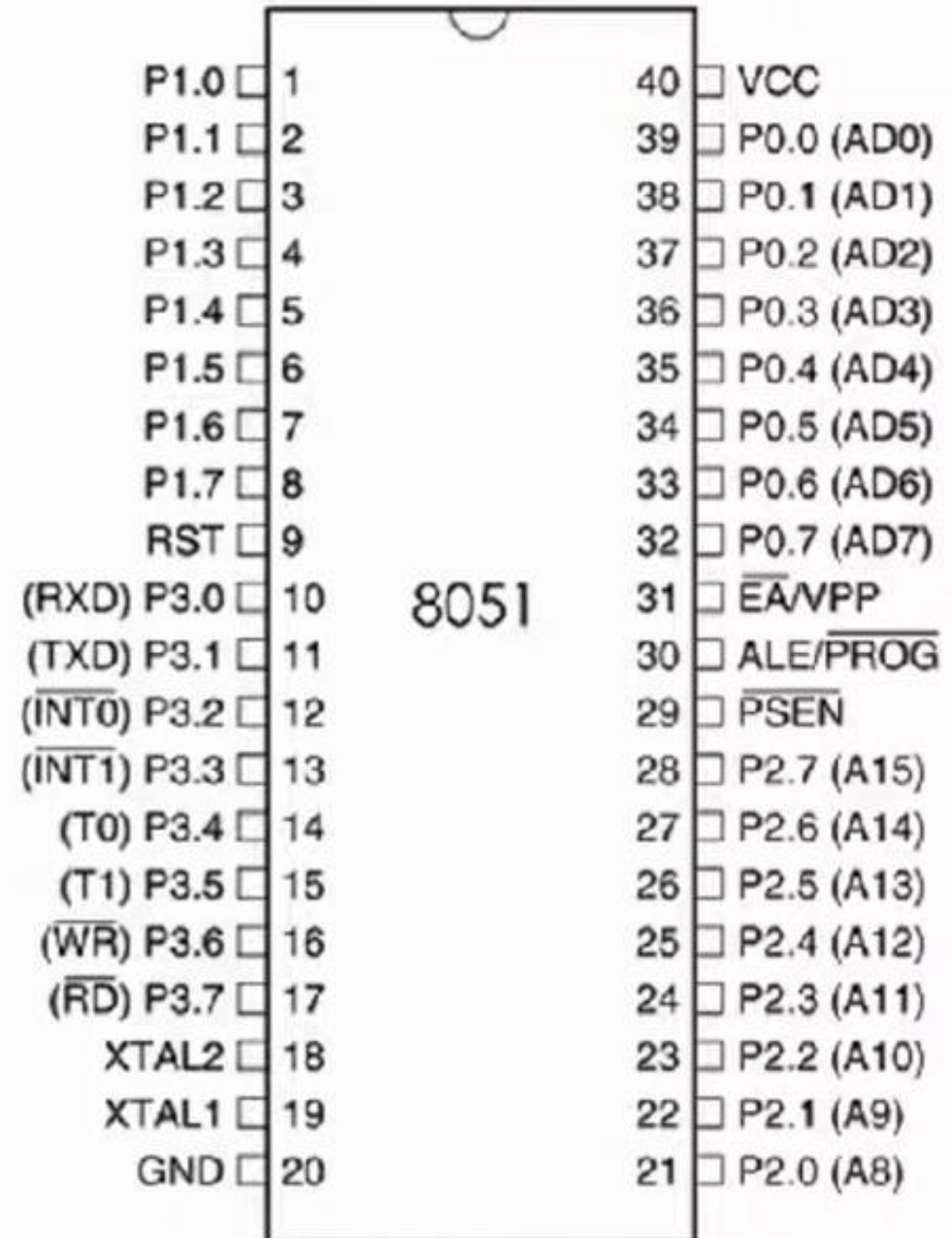
Lecture 2

Pin Diagram and Description

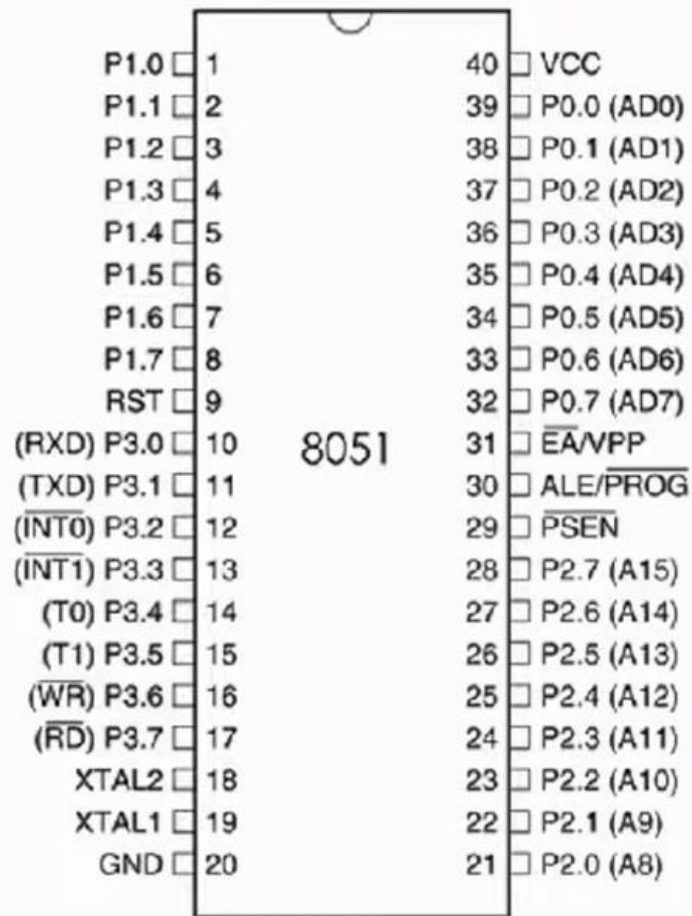
Feature of Microcontroller

- ☐ 8051 is complete computer system built on one Chip. It has CPU, RAM, ROM, Serial port, Parallel Port, Interrupt, Timer on single chip.
- ☐ 8051 operates at 12MHz clock frequency.
- ☐ 8051 has 8 bits of ALU.
- ☐ 8051 has 8 bits of Data lines.
- ☐ 8051 follows Harvard architecture. {So, it has separate memory for program and data storage.}
- ☐ 8051 has 4KB of internal ROM for program storage.
- ☐ 8051 has 128 bytes of RAM for data storage.
- ☐ 8051 has Four 8 bits of IO ports. {This ports can be used for interfacing of peripherals like Keyboard, display, stepper motor, LEDs & switches etc}
- ☐ 8051 has a serial port synchronous and asynchronous communication.
- ☐ 8051 has two 16 bits 'UP' timers. {which can be used for delay generation.}
- ☐ 8051 has Five interrupts operating at priority levels.
- ☐ 8051 has two power saving modes. {Idle mode and Power down mode}
- ☐ 8051 has 16 bits of address bus which can be used to interface external memory of RAM or ROM. {64KB can be interfaced with 16 address lines.}

Pin Diagram of 8051



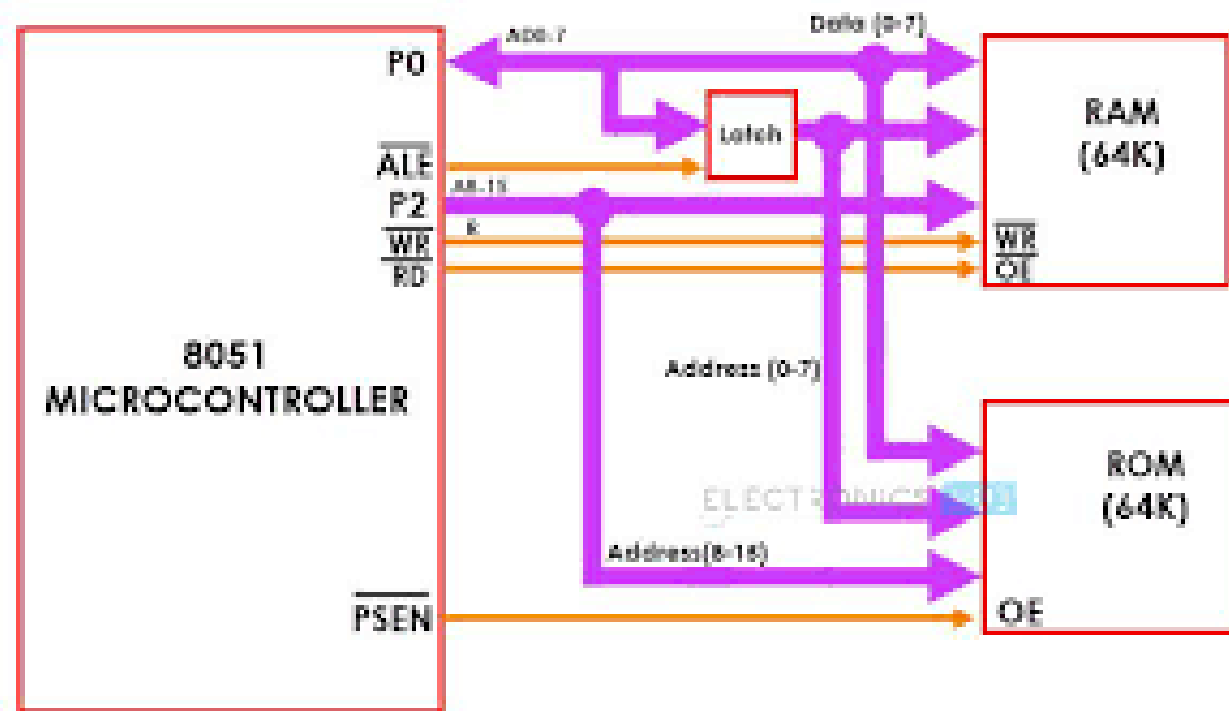
Pin Description



- Intel 8051 is a 40 pin IC available in Dual-In line package (DIP), which is an electronic component package with a rectangular housing and two parallel rows of electrical connecting pins.
- Requires a single power supply of +5V
- VCC and GND
- P0 Port 0 is multiplexed address and data bus
- P1 Port 1
- P2 input/output ports and it can allow 16-bit data bus
- P3 Port 3 and each pin has a special function
- XTAL1 and XTAL2 is External Crystal
- RST ReSeT pin
- EA/Vpp Enable Address pin is used to supply programming voltage 12V to EPROM/ROM
- Address latch enable It is used to demultiplex address and data bus. It separated A0-A7 address from AD0-AD7

Pin Diagram

- PSEN is used to enable external program memory(ROM) It is connected to (OE)of external ROM



PSW Program Status Word Register of 8051

- Format of Program Status Word Register

D7	D6	D5	D4	D3	D2	D1	D0
CY	AC	F0	RS1	RS0	OV	-	P

D7	D6	D5	D4	D3	D2	D1	D0
CY	AC	F0	RS1	RS0	OV	-	P

- CY Carry Flag

$$\begin{array}{r}
 11110101 \\
 + 10001010 \\
 \hline
 10111111 \\
 \text{CY}
 \end{array}$$

D7	D6	D5	D4	D3	D2	D1	D0
CY	AC	F0	RS1	RS0	OV	-	P

- AC Auxiliary Carry Flag

$$\begin{array}{r}
 \text{CY} \\
 \textcolor{red}{1} \\
 1101 \quad \boxed{1}101 \\
 + \quad 0000 \quad \boxed{1}010 \\
 \hline
 0111 \quad 0111
 \end{array}$$

D7	D6	D5	D4	D3	D2	D1	D0
CY	AC	F0	RS1	RS0	OV	-	P

- F0 Frag 0 available to user for general purpose

D7	D6	D5	D4	D3	D2	D1	D0
CY	AC	F0	RS1	RS0	OV	-	P

- RS1 and RS0 is 2 bits of Register bank selector (bit 1 and 0) of internal RAM for pointing a sector of internal memory

RS1	RS0	Register Bank
0	0	0
0	1	1
1	0	2
1	1	3

D7	D6	D5	D4	D3	D2	D1	D0
CY	AC	F0	RS1	RS0	OV	-	P

- OV Overflow flag is set to 1 when the result of operation is greater than 8 bits or can be stored in 8 bit registers
- P Parity Flag set or cleared by hardware during instruction cycle to indicate even($P=0$) or odd($P=1$) of 1 bit in accumulator

1 0 0 1 0 1 0 0	P=1
1 1 0 1 0 1 0 0	P=0